

Tanuj Haresh Sidhwa

B.Tech in Mechanical Engineering (Minor: Astronomy)

Indian Institute of Technology Indore

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[Portfolio](#) | [LinkedIn](#) | [GitHub](#)

Research Interests: Computational Fluid Dynamics, Combustion & Propulsion, Electrohydrodynamics, Aerospace Systems, Physics-Informed Neural Networks, Reduced Order Modelling

EDUCATION

• Indian Institute of Technology, Indore

2023 - 2027

– B.Tech in Mechanical Engineering, CGPA: 9.47/10

Indore, India

– Minor in Astronomy, CGPA: 9.75/10

TECHNICAL SKILLS

• **Simulation & Aerodynamic Analysis:** ANSYS Mechanical, ANSYS Fluent, XFLR5, XFOIL, OpenFOAM

• **CAD & Design:** CATIA V5, Fusion 360, ANSYS SpaceClaim

• **Scientific Computing & Machine Learning:** Python, MATLAB, C++, NumPy, SciPy, PyTorch

EXPERIENCE

• Liquid Propulsion Systems Centre (LPSC), ISRO

Jun. 2026 - Jul. 2026

Research Intern / Guide: Vijay Kumar, Scientist/Engineer 'SG'

Bengaluru, India

- Developed and improved an OpenFOAM-based electrohydrodynamic solver for cone-jet electrosprays through boundary-condition corrections, mesh refinement, interface treatment, and storage optimization.
- Validated the computational framework against LPSC experimental cone-jet observations, reproducing stable Taylor-cone formation and physically consistent electrospray behavior for subsequent parametric studies.
- Investigated the effects of voltage, flow rate, electrode spacing, and emitter geometry, identifying electric-field strength and emitter design as primary drivers of emitted current and electrospray performance.

• IASc-INSa-NASI Summer Research Fellowship

May 2025 - Jul. 2025

Research Intern / Dr Abhijit Kushari, Advanced Propulsion Laboratory

Indian Institute of Technology, Kanpur

- Conducted static structural analysis of serpentine nozzles in ANSYS, evaluating deformation behavior under fluid-induced and inertial loading conditions relevant to propulsion and high-speed flow applications.
- Performed fatigue, modal, and harmonic analyses to estimate component lifespan, identify resonance conditions through Campbell diagrams, and evaluate structural integrity under complex dynamic excitation environments.
- Studied plasticity model implementation using bilinear isotropic hardening models, while developing proficiency in CATIA V5 and conducting preliminary fluid-structure interaction studies supported by literature review.

PROJECTS

• Combustion and Exfoliation of Inorganic Gellant-Laden Ethanol Droplets

Aug. 2026 - Present

B.Tech Project under supervision of Dr. Ankur Miglani

- Studying combustion and exfoliation behavior of gellant-laden ethanol droplets through controlled experiments.
- Investigating effects of gellant composition and concentration on droplet morphology and combustion behavior.

• Physics-Informed Neural Network for Diesel Engine

Jan. 2026 - Present

Prof. Devendra Deshmukh, Department of Mechanical Engineering

- Developed a Fourier PINN in PyTorch to predict in-cylinder pressure from Sandia ECN diesel combustion data.
- Embedded thermodynamic constraints and regularization, achieving <7% error on unseen injection profiles.

• Computational Modeling of Propellant Sloshing Dynamics in Spacecraft Tanks

Aug. 2026 - Present

B.Tech Minor Project under supervision of Dr. Bhargav Vaidya

- Developing an OpenFOAM-based computational model for propellant sloshing in axisymmetric spacecraft tanks using potential-flow theory.
- Deriving equivalent pendulum and spring-mass models to characterize slosh dynamics across tank geometries and fill levels.

- **Numerical Design and Performance Mapping of a Micro Cold-Gas Thruster** Jan. 2026 - Apr. 2026
Course Project: Computational Fluids and Structures (AA374), IIT Indore [GitHub](#)
 - Developed a quasi-1D finite-volume solver for compressible nozzle flows using the Euler equations.
 - Implemented Lax–Friedrichs, HLLC, and MUSCL–HLLC schemes and validated against theory and ANSYS.
- **Powered Lift Wing Design for STOL Applications** Oct. 2025 - Dec. 2025
InterIIT Tech Meet 14.0: High Prep – LAT Aerospace
 - Executed CFD-driven aerodynamic design of a powered-lift STOL wing, including high-lift configuration studies, for InterIIT Tech Meet 14.0.
 - Optimized aerodynamic performance through ANSYS Fluent studies, balancing lift, drag, and stability to achieve $C_{L,max} = 6.81$.
- **Assessment of Indoor Air Quality in Student Hostel** Sept. 2024 - Oct. 2025
Prof. Ritunesh Kumar, Department of Mechanical Engineering [GitHub](#)
 - Developed an Arduino-based CO₂ monitoring system for controlled occupancy and ventilation experiments.
 - Analyzed transient CO₂ buildup trends; work resulted in a first-author paper at INCOTHERM 2025.
- **CubeSat Project** Oct. 2024 - Nov. 2024
CAE Club & Astronomy Club, IIT Indore [GitHub](#)
 - Managed CubeSat structural design and cross-domain coordination within a multidisciplinary team.
 - Designed the chassis and performed structural, modal, and thermal analyses under launch loads.

PUBLICATIONS & CONFERENCES

- **1st International Conference on Thermofluids Engineering (INCOTHERM-2025)** Oct. 2025
First Author IIT (ISM) Dhanbad
 - First-author paper, “*Experimental Investigations on Effects of Student Occupancy Rate in a Student Hostel Room on Indoor Air Quality*,” accepted and presented; proceedings under publication process.
- **Students’ Academic Conference (SAC)** Dec. 2025
Presenter Inter IIT Tech Meet 14.0, IIT Patna
 - Presented “*An Innovative Laser Decal Transfer of ZnO Ceramic in LIG for Advanced Hybrid Nanogenerator Applications*” on behalf of the primary author.

RELEVANT COURSEWORK

- **Mathematics:** Linear Algebra, Differential Equations, Multivariable Calculus, Real Analysis, Complex Analysis, Numerical Methods, Calculus of Variation
- **Mechanical Engineering:** Fluid Mechanics, Thermodynamics, Heat Transfer, Solid Mechanics, Instrumentation and Control Systems, Design of Machine Elements, Principles of Industrial Engineering
- **Technical Electives:** Computational Fluids and Structures, Introduction to Finite Element Methods, Fundamentals of Vibrations, Fundamentals of Acoustics, Introduction to Robotics
- **Astronomy Minor:** Flight Mechanics and Classical Control, High Energy Astrophysics and Transient Sky, IoT for Space Applications, Introduction to Astronomy

ACHIEVEMENTS

- **Reliance Foundation Undergraduate Scholar**, Awarded to meritorious students nationwide 2024
- **Bronze Medal: IITISoC’24**, Design & Modelling, Material Selection, Design and FEA of CubeSat 2024
- **6th Position:** InterIIT Techmeet 14.0, Lat Aerospace HighPrep PS, Powered Lift Wing Design 2025
- **AIR 144:** GATE Aerospace Engineering (AE), Secured All India Rank 144 in GATE AE 2026 2026
- **AIR 1224:** GATE Mechanical Engineering (ME), Secured All India Rank 1224 in GATE ME 2026 2026
- **Awarded AP grade in AA374: Computational Fluids and Structures**, for exceptional performance, securing the top rank in the class. 2026

POSITIONS OF RESPONSIBILITY

- **President**, Mechanical Engineering Student Association, IIT Indore Apr. 2025 - Apr. 2026
 - **Team Lead**, High Prep LAT Aerospace, InterIIT Techmeet 14.0 Oct. 2025 - Dec. 2025
 - **Project Lead**, CubeSat Project, IIT Indore Oct. 2024 - Nov. 2024
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